

```
usepackage[a4paper,margin=0.65in]geometry
```

MATHEMATICAL PAPERS.

LONDON: C. J. CLAY AND SONS,
CAMBRIDGE UNIVERSITY PRESS WAREHOUSE,
AVE MARIA LANE.
Glasgow: 263, ARGYLE STREET.

Leipzig: F. A. BROCKHAUS.
New York: THE MACMILLAN COMPANY.

THE COLLECTED
MATHEMATICAL PAPERS

OF

ARTHUR CAYLEY, Sc.D., F.R.S.,
LATE SADLERIAN PROFESSOR OF PURE MATHEMATICS IN THE UNIVERSITY
OF CAMBRIDGE.

VOL. XII.

CAMBRIDGE:
AT THE UNIVERSITY PRESS.
1897.

[*All Rights reserved.*]

CAMBRIDGE:
PRINTED BY J. AND C. F. CLAY,
AT THE UNIVERSITY PRESS.

ADVERTISEMENT.

THE present volume contains 89 papers, numbered 799 to 887, published for the most part in the years 1883 to 1889.

The Table for the twelve volumes is

Vol.	I.	Numbers	1 to 100,
"	II.	"	101 to 158,
"	III.	"	159 to 222,
"	IV.	"	223 to 299,
"	V.	"	300 to 383,
"	VI.	"	384 to 416,
"	VII.	"	417 to 485,
"	VIII.	"	486 to 555,
"	IX.	"	556 to 629,
"	X.	"	630 to 705,
"	XI.	"	706 to 798,
"	XII.	"	799 to 887.

A. R. FORSYTH.

19 May, 1897.

CONTENTS.

[An Asterisk means that the paper is not printed in full.]

	PAGE
799. <i>On curvilinear coordinates</i>	1
Quart. Math. Journ., t. XIX. (1883), pp. 1—22	
800. <i>Note on the standard solutions of a system of linear equations</i>	19
Quart. Math. Journ., t. XIX. (1883), pp. 38—40	
801. <i>On seminvariants</i>	22
Quart. Math. Journ., t. XIX. (1883), pp. 131—138	
802. <i>Note on Captain MacMahon's paper "On the differential equation $X^{-\frac{1}{3}}dx + Y^{-\frac{1}{3}}dy + Z^{-\frac{1}{3}}dz = 0$"</i>	30
Quart. Math. Journ., t. XIX. (1883), pp. 182—184	
803. <i>On Mr Anglin's formula for the successive powers of the root of an algebraical equation</i>	33
Quart. Math. Journ., t. XIX. (1883), pp. 223, 224	
804. <i>On the elliptic-function solution of the equation $x^3 + y^3 - 1 = 0$</i>	35
Camb. Phil. Soc. Proc., t. IV. (1883), pp. 106—109	
805. <i>Note on Abel's theorem</i>	38
Camb. Phil. Soc. Proc., t. IV. (1883), pp. 119—122	
806. <i>Determination of the order of a surface</i>	42
Messenger of Mathematics, t. XII. (1883), pp. 29—32	
807. <i>A proof of Wilson's theorem</i>	45
Messenger of Mathematics, t. XII. (1883), p. 41	

	PAGE
808. <i>Note on a form of the modular equation in the transformation of the third order</i>	46
Messenger of Mathematics, t. XII. (1883), pp. 173, 174	
809. <i>Schröter's construction of the regular pentagon</i>	47
Messenger of Mathematics, t. XII. (1883), p. 177	
810. <i>Note on a system of equations</i>	48
Messenger of Mathematics, t. XII. (1883), pp. 191, 192	
811. <i>On the linear transformation of the theta-functions</i>	50
Messenger of Mathematics, t. XIII. (1884), pp. 54—60	
812. <i>On Archimedes' theorem for the surface of a cylinder</i>	56
Messenger of Mathematics, t. XIII. (1884), pp. 107, 108	
813. [<i>Note on Mr Griffiths' paper "On a deduction from the elliptic-integral formula $y = \sin(A + B + C + \dots)$"</i>]	58
Proc. Lond. Math. Soc., t. XV. (1884), p. 81	
814. <i>On double algebra</i>	60
Proc. Lond. Math. Soc., t. XV. (1884), pp. 185—197	
815. <i>The binomial equation $x^p - 1 = 0$; quinquisection. Second part</i>	72
Proc. Lond. Math. Soc., t. XVI. (1885), pp. 61—63	
816. <i>On the bitangents of a plane quartic</i>	74
Crelle's Journal der Mathem., t. XCIV. (1883), pp. 93—115; Camb. Phil. Soc. Proc., t. IV. (1883), p. 321	
817. <i>On the sixteen-nodal quartic surface</i>	95
Crelle's Journal der Mathem., t. XCIV. (1883), pp. 270—272	
*818. <i>Note on hyperelliptic integrals of the first order</i>	98
Crelle's Journal der Mathem., t. XCVIII. (1885), pp. 95, 96	
819. <i>On two cases of the quadric transformation between two planes</i>	100
Johns Hopkins University Circulars, No. 13 (1882), pp. 178, 179	
820. <i>On a problem of analytical geometry</i>	102
Johns Hopkins University Circulars, No. 15 (1882), p. 209	

	PAGE
821. <i>On the geometrical representation of an equation between two variables</i> .	104
Johns Hopkins University Circulars, No. 15 (1882), p. 210	
822. <i>On associative imaginaries</i>	105
Johns Hopkins University Circulars, No. 15 (1882), pp. 211, 212	
823. <i>On the geometrical interpretation of certain formulæ in elliptic functions</i>	107
Johns Hopkins University Circulars, No. 17 (1882), p. 238	
824. <i>Note on the formulæ of trigonometry</i>	108
Johns Hopkins University Circulars, No. 17 (1882), p. 241	
825. <i>A memoir on the Abelian and Theta Functions</i>	109
Chapters I to III, American Journal of Mathematics, t. v. (1882), pp. 137—179; Chapters IV to VII, ib., t. vii. (1885), pp. 101—167	
826. <i>Note on a partition series</i>	217
American Journal of Mathematics, t. vi. (1884), pp. 63, 64	
827. <i>On the non-Euclidian plane geometry</i>	220
Proc. Roy. Soc., t. XXXVII. (1884), pp. 82—102	
828. <i>A memoir on seminvariants</i>	239
American Journal of Mathematics, t. vii. (1885), pp. 1—25	
829. <i>Tables of the symmetric functions of the roots, to the degree 10, for the form</i> $1 + bx + \frac{cx^2}{1.2} + \dots = (1 - \alpha x)(1 - \beta x)(1 - \gamma x) \dots$	263
American Journal of Mathematics, t. vii. (1885), pp. 47—56	
830. <i>Non-unitary partition tables</i>	273
American Journal of Mathematics, t. vii. (1885), pp. 57, 58	
831. <i>Seminvariant tables</i>	275
American Journal of Mathematics, t. vii. (1885), pp. 59—73	
832. <i>Note on an apparent difficulty in the theory of curves, when the coordinates of a point are given as functions of a variable parameter</i>	290
Messenger of Mathematics, t. xiv. (1885), pp. 12—14	

	PAGE
833. <i>On a formula in elliptic functions</i>	292
Messenger of Mathematics, t. XIV. (1885), pp. 21, 22	
834. <i>On the addition of the elliptic functions</i>	294
Messenger of Mathematics, t. XIV. (1885), pp. 56—61	
835. <i>On Cardan's solution of a cubic equation</i>	299
Messenger of Mathematics, t. XIV. (1885), pp. 96, 97	
836. <i>On the quaternion equation $qQ - Qq' = 0$</i>	300
Messenger of Mathematics, t. XIV. (1885), pp. 108—112	
837. <i>On the so-called D'Alembert-Carnot geometrical paradox</i>	305
Messenger of Mathematics, t. XIV. (1885), pp. 113, 114	
838. <i>On the twisted cubics upon a quadric surface</i>	307
Messenger of Mathematics, t. XIV. (1885), pp. 129—132	
839. <i>On the matrical equation $qQ - Qq' = Q$</i>	311
Messenger of Mathematics, t. XIV. (1885), pp. 176—178	
840. <i>On Mascheroni's geometry of the compass</i>	314
Messenger of Mathematics, t. XIV. (1885), pp. 179—181	
841. <i>On a differential operator</i>	318
Messenger of Mathematics, t. XIV. (1885), pp. 190, 191	
842. <i>On the value of $\tan(\sin \theta) - \sin(\tan \theta)$</i>	319
Messenger of Mathematics, t. XIV. (1885), pp. 191, 192	
843. <i>On the quadri-quadric curve in connexion with the theory of elliptic functions</i>	321
Mathematische Annalen, t. XXV. (1885), pp. 152—156	
844. <i>On a theorem relating to seminvariants</i>	326
Quart. Math. Journ., t. XX. (1885), pp. 212, 213	
845. <i>On the orthomorphosis of the circle into the parabola</i>	328
Quart. Math. Journ., t. XX. (1885), pp. 213—220	
846. <i>A verification in regard to the linear transformation of the theta-functions</i>	337
Quart. Math. Journ., t. XXI. (1886), pp. 77—84	

	PAGE
847. <i>On the theory of seminvariants</i>	344
Quart. Math. Journ., t. XXI. (1886), pp. 92—107	
848. <i>On the transformation of the double theta-functions</i>	358
Quart. Math. Journ., t. XXI. (1886), pp. 142—178	
849. <i>On the invariants of a linear differential equation</i>	390
Quart. Math. Journ., t. XXI. (1886), pp. 257—261	
850. <i>On linear differential equations</i>	394
Quart. Math. Journ., t. XXI. (1886), pp. 321—331	
851. <i>On linear differential equations: the theory of decomposition</i>	403
Quart. Math. Journ., t. XXI. (1886), pp. 331—335	
852. <i>Note sur le mémoire de M. Picard “Sur les intégrales de différentielles totales algébriques de première espèce”</i>	408
Bull. des Sciences Math., 2 ^{me} Sér., t. X. (1886), pp. 75—78	
853. <i>Note on a formula for $\Delta^n 0^i / n^i$ when n, i are very large numbers</i>	412
Proc. Roy. Soc. Edin., t. XIV. (1887), pp. 149—153	
854. <i>An algebraical transformation</i>	416
Messenger of Mathematics, t. XV. (1886), pp. 58, 59	
855. <i>Solution of $(a, b, c, d)(x, y)^3 = (a', b', c', d')(x', y')^3$</i>	418
Messenger of Mathematics, t. XV. (1886), pp. 59—61	
856. <i>Note on a cubic equation</i>	421
Messenger of Mathematics, t. XV. (1886), pp. 62—64	
857. <i>Analytical geometrical note on the conic</i>	424
Messenger of Mathematics, t. XV. (1886), p. 192	
858. <i>Comparison of the Weierstrassian and Jacobian elliptic functions</i>	425
Messenger of Mathematics, t. XVI. (1887), pp. 129—132	
859. <i>On the complex of lines which meet a unicursal quartic curve</i>	428
Proc. Lond. Math. Soc., t. XVII. (1886), pp. 232—238	

b 2

	PAGE
860. <i>On Briot and Bouquet's theory of the differential equation $F\left(u, \frac{du}{dz}\right) = 0$</i>	432
Proc. Lond. Math. Soc., t. XVIII. (1887), pp. 314—324	
861. <i>Note on a formula relating to the zero-value of a theta-function</i>	442
Crelle's Journal der Mathem., t. C. (1887), pp. 87, 88	
862. <i>Note on the theory of linear differential equations</i>	444
Crelle's Journal der Mathem., t. C. (1887), pp. 286—295	
863. <i>Note on the theory of linear differential equations</i>	453
Crelle's Journal der Mathem., t. CI. (1887), pp. 209—213	
864. <i>On Rudio's inverse centro-surface</i>	457
Quart. Math. Journ., t. XXII. (1887), pp. 156—158	
865. <i>On multiple algebra</i>	459
Quart. Math. Journ., t. XXII. (1887), pp. 270—308	
866. <i>Note on Kiepert's L-equations, in the transformation of elliptic functions</i>	490
Mathematische Annalen, t. XXX. (1887), pp. 75—77	
867. <i>Note on the Jacobian sextic equation</i>	493
Mathematische Annalen, t. XXX. (1887), pp. 78—84	
868. <i>On the intersection of curves</i>	500
Mathematische Annalen, t. XXX. (1887), pp. 85—90	
869. <i>On the transformation of elliptic functions</i>	505
American Journal of Mathematics, t. IX. (1887), pp. 193—224	
870. <i>On the transformation of elliptic functions (sequel)</i>	535
American Journal of Mathematics, t. X. (1888), pp. 71—93	
871. <i>A case of complex multiplication with imaginary modulus arising out of the cubic transformation in elliptic functions</i>	556
Proc. Lond. Math. Soc., t. XIX. (1888), pp. 300, 301	
*872. <i>On the finite number of the covariants of a binary quantic</i>	558
Mathematische Annalen, t. XXXIV. (1889), pp. 319, 320	

	PAGE
873. <i>System of equations for three circles which cut each other at given angles</i> 559 Messenger of Mathematics, t. XVII. (1888), pp. 18—21	
874. <i>Note on the Legendrian coefficients of the second kind</i> 562 Messenger of Mathematics, t. XVII. (1888), pp. 21—23	
875. <i>On the system of three circles which cut each other at given angles and have their centres in a line</i> 564 Messenger of Mathematics, t. XVII. (1888), pp. 60—69	
876. <i>On systems of rays</i> 571 Messenger of Mathematics, t. XVII. (1888), pp. 73—78	
877. <i>Note on the two relations connecting the distances of four points on a circle</i> 576 Messenger of Mathematics, t. XVII. (1888), pp. 94, 95	
878. <i>Note on the anharmonic ratio equation</i> 578 Messenger of Mathematics, t. XVII. (1888), pp. 95, 96	
879. <i>Note on the differential equation $\frac{dx}{\sqrt{1-x^2}} + \frac{dy}{\sqrt{1-y^2}} = 0$</i> 580 Messenger of Mathematics, t. XVIII. (1889), p. 90	
880. <i>Note on the relation between the distance of five points in space</i> 581 Messenger of Mathematics, t. XVIII. (1889), pp. 100—102	
881. <i>On Hermite's H-product theorem</i> 584 Messenger of Mathematics, t. XVIII. (1889), pp. 104—107	
882. <i>A correspondence of confocal Cartesians with the right lines of a hyperboloid</i> 587 Messenger of Mathematics, t. XVIII. (1889), pp. 128—130	
883. <i>Analytical formulæ in regard to an octad of points</i> 590 Messenger of Mathematics, t. XVIII. (1889), pp. 149—152	
884. <i>Note sur les surfaces minima et le théorème de Joachimsthal</i> 594 Comptes Rendus, t. CVI. (1888), pp. 995, 996	

	PAGE
885. <i>On the Diophantine relation, $y^2 + y'^2 = \text{Square}$</i>	596
Proc. Lond. Math. Soc., t. XX. (1889), pp. 122—127	
886. <i>On the surfaces with plane or spherical curves of curvature</i>	601
American Journal of Mathematics, t. XI. (1889), pp. 71—98; pp. 293—306	
887. <i>On the theory of groups</i>	639
American Journal of Mathematics, t. XI. (1889), pp. 139—157	

CLASSIFICATION.

ANALYSIS.

Equations, 801, 803, 835, 855, 856, 867, 878.
Theory of numbers, 807, 815.
Groups, 887.
Quaternions, 836.
Matrices, 839.
Trigonometry, 824, 842.
Symmetric functions, 829, 841.
Partitions, 826, 830.
Invariants and covariants, 800, 801, 828, 831, 844, 847, *872.
Multiple Algebra, 814, 822, 865.
Integral calculus, 852.
Legendre's coefficients, 874.
Finite differences, 853.
Differential equations (general), 802, 860, 879, 885.
Differential equations (linear), 849, 850, 851, 862, 863.
Elliptic functions, 804, 813, 823, 833, 834, 843, 858, 881.
Transformation of elliptic functions, 808, 854, 866, 869, 870, 871.
Abel's theorem, 805.
Abelian functions, 818, 825.
Theta functions, 811, 825, 846, 848, 861.

GEOMETRY.

Properties of circles, 873, 875, 877.
Non-Euclidian geometry, 827.
General theory of curves, 820, 832, 868.
Correspondence, 819, 821.
Orthomorphosis, 821, 845.
Plane Quartics, 816.
Groups of points, 880, 883.
General theory of surfaces, 799, 806.
Quadrics, 882.
Quartic surfaces, 817.
Centro-surfaces, 864.
Minimal surfaces, 884.
Rays, 876.
Curves of curvature, 886.
Twisted cubic curves, 838.
Twisted quartic curves, 843, 859.

MISCELLANEOUS, 809, 810, 812, 837, 840, 857.

799.

ON CURVILINEAR COORDINATES.

[*From the Quarterly Journal of Pure and Applied Mathematics*, vol. XIX. (1883), pp. 1—22.]

THE present memoir is based upon Mr Warren's "Exercises in Curvilinear and Normal Coordinates," *Camb. Phil. Trans.* t. XII. (1877), pp. 455—502, and has for a principal object the establishment of the six differential equations of the second order corresponding to his six equations for normal coordinates: but the notation is different; the results are more general, inasmuch as I use throughout general curvilinear coordinates instead of his normal coordinates; and as regards my six equations for general curvilinear coordinates, the terms containing differential coefficients of the first order are presented under a different form.

If the position of a point in space is determined by the rectangular coordinates x, y, z ; then p, q, r being each of them a given function of x, y, z , we have conversely x, y, z , each of them a given function of p, q, r , which are thus in effect coordinates serving to determine the position of the point, and are called curvilinear coordinates.

But it is not in the first instance necessary to regard x, y, z as rectangular coordinates, or even as Cartesian coordinates at all; we are simply concerned with the two sets of variables x, y, z and p, q, r , each variable of the one set being a given function of the variables of the other set; and, in particular, the x, y, z are regarded as being each of them a given function of the p, q, r .

Except as regards the symbols ξ, η, ζ presently mentioned, the suffixes 1, 2, 3 refer to the variables p, q, r respectively, and are used to denote differentiations in regard to these variables, viz.

$$x_1, x_2, x_3, x_{11}, x_{22}, x_{33}, \dots$$

are written to denote

$$\frac{dx}{dp}, \frac{dx}{dq}, \frac{dx}{dr}, \frac{d^2x}{dp^2}, \frac{d^2x}{dq^2}, \frac{d^2x}{dr^2}, \text{ \&c.},$$

and so in other cases; in particular,

$$x_1, x_2, x_3 \text{ denote } \frac{dx}{dp}, \frac{dx}{dq}, \frac{dx}{dr},$$

$$y_1, y_2, y_3 \quad , , \quad \frac{dy}{dp}, \frac{dy}{dq}, \frac{dy}{dr},$$

$$z_1, z_2, z_3 \quad , , \quad \frac{dz}{dp}, \frac{dz}{dq}, \frac{dz}{dr}.$$

The minors formed with these differential coefficients are denoted by suffixed letters ξ, η, ζ , thus

$$\xi_1, \xi_2, \xi_3 \text{ denote } y_2z_3 - y_3z_2, y_3z_1 - y_1z_3, y_1z_2 - y_2z_1,$$

$$\eta_1, \eta_2, \eta_3 \quad , , \quad z_2x_3 - z_3x_2, z_3x_1 - z_1x_3, z_1x_2 - z_2x_1,$$

$$\zeta_1, \zeta_2, \zeta_3 \quad , , \quad x_2y_3 - x_3y_2, x_3y_1 - x_1y_3, x_1y_2 - x_2y_1,$$

so that, as regards these letters ξ, η, ζ , the suffixes do not denote differentiations.

The determinant

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \text{ is put } = L;$$

and the symbols $(a, b, c, f, g, h), (A, B, C, F, G, H)$ are defined as follows:

$$\begin{aligned} a &= x_1^2 + y_1^2 + z_1^2, & A &= \xi_1^2 + \eta_1^2 + \zeta_1^2, \\ b &= x_2^2 + y_2^2 + z_2^2, & B &= \xi_2^2 + \eta_2^2 + \zeta_2^2, \\ c &= x_3^2 + y_3^2 + z_3^2, & C &= \xi_3^2 + \eta_3^2 + \zeta_3^2, \\ f &= x_2x_3 + y_2y_3 + z_2z_3, & F &= \xi_2\xi_3 + \eta_2\eta_3 + \zeta_2\zeta_3, \\ g &= x_3x_1 + y_3y_1 + z_3z_1, & G &= \xi_3\xi_1 + \eta_3\eta_1 + \zeta_3\zeta_1, \\ h &= x_1x_2 + y_1y_2 + z_1z_2, & H &= \xi_1\xi_2 + \eta_1\eta_2 + \zeta_1\zeta_2. \end{aligned}$$

We have then, further,

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = L, \quad \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = L^2,$$

$$\begin{vmatrix} \xi_1 & \eta_1 & \zeta_1 \\ \xi_2 & \eta_2 & \zeta_2 \\ \xi_3 & \eta_3 & \zeta_3 \end{vmatrix} = L^2, \quad \begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix} = \frac{1}{L^2};$$

$$(A, B, C, F, G, H) = (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch),$$

$$L^2(a, b, c, f, g, h) = (BC - F^2, CA - G^2, AB - H^2, GH - AF, HF - BG, FG - CH),$$

which equations are at once proved, and are fundamental ones in the theory.

It is convenient to add that we have

$$\begin{vmatrix} \frac{dp}{dx} & \frac{dp}{dy} & \frac{dp}{dz} \\ \frac{dq}{dx} & \frac{dq}{dy} & \frac{dq}{dz} \\ \frac{dr}{dx} & \frac{dr}{dy} & \frac{dr}{dz} \end{vmatrix} = \frac{1}{L} \begin{vmatrix} \xi_1 & \eta_1 & \zeta_1 \\ \xi_2 & \eta_2 & \zeta_2 \\ \xi_3 & \eta_3 & \zeta_3 \end{vmatrix},$$

that is, $\frac{dp}{dx} = \frac{1}{L} \xi_1$, &c.; and, further,

$$dx^2 + dy^2 + dz^2 = (a, b, c, f, g, h \mid dp, dq, dr)^2,$$

$$dp^2 + dq^2 + dr^2 = \frac{1}{L^2} (A, B, C, F, G, H \mid dx, dy, dz)^2.$$

Differentiating the values of a, b, c, f, g, h , we have $\frac{1}{2}a_1 = x_1x_{11} + y_1y_{11} + z_1z_{11}$, which may be written in the abbreviated form $\frac{1}{2}a_1 = 1.11$; similarly

$$x_1x_{12} + x_2x_{11} + y_1y_{12} + y_2y_{11} + z_1z_{12} + z_2z_{11},$$

which in like manner may be written $f_1 = 2.13 + 3.12$, and so in other cases; observe that, in the duad part of any symbol, the order of the numbers is immaterial, $2.13 = 2.31$. The whole system of equations is

$$\begin{array}{lll} \frac{1}{2}a_1 = 1.11, & \frac{1}{2}a_2 = 1.12, & \frac{1}{2}a_3 = 1.13, \\ \frac{1}{2}b_1 = 2.12, & \frac{1}{2}b_2 = 2.22, & \frac{1}{2}b_3 = 2.23, \\ \frac{1}{2}c_1 = 3.13, & \frac{1}{2}c_2 = 3.23, & \frac{1}{2}c_3 = 3.33, \\ f_1 = 2.13 + 3.12, & f_2 = 2.23 + 3.22, & f_3 = 2.33 + 3.23, \\ g_1 = 3.11 + 1.13, & g_2 = 3.12 + 1.23, & g_3 = 3.13 + 1.33, \\ h_1 = 1.12 + 2.11, & h_2 = 1.22 + 2.12, & h_3 = 1.23 + 2.13. \end{array}$$

These may also be written

$$\begin{array}{lll} 1.11 = \frac{1}{2}a_1, & 1.22 = h_2 - \frac{1}{2}b_1, & 1.33 = g_3 - \frac{1}{2}c_1, \\ 2.11 = h_1 - \frac{1}{2}a_2, & 2.22 = \frac{1}{2}b_2, & 2.33 = f_3 - \frac{1}{2}c_2, \\ 3.11 = g_1 - \frac{1}{2}a_3, & 3.22 = f_2 - \frac{1}{2}b_3, & 3.33 = \frac{1}{2}c_3, \\ 1.23 = \frac{1}{2}(-f_1 + g_2 + h_3), & 1.31 = \frac{1}{2}a_3, & 1.12 = \frac{1}{2}a_2, \\ 2.23 = \frac{1}{2}b_3, & 2.31 = \frac{1}{2}(f_1 - g_2 + h_3), & 2.12 = \frac{1}{2}b_1, \\ 3.23 = \frac{1}{2}c_2, & 3.31 = \frac{1}{2}c_1, & 3.12 = \frac{1}{2}(f_1 + g_2 - h_3). \end{array}$$

It is to be observed that we can, from each system of three equations, express a set of second differential coefficients of the x, y, z in terms of the first differential

coefficients of the a, b, c, f, g, h; thus the three equations containing 11, written at length, are

$$\begin{aligned} x_1 x_{11} + y_1 y_{11} + z_1 z_{11} &= \frac{1}{2} a_1, \\ x_2 + y_2 + z_2 &= h_1 - \frac{1}{2} a_2, \\ x_3 + y_3 + z_3 &= g_1 - \frac{1}{2} a_3, \end{aligned}$$

three linear equations for the determination of x_{11} , y_{11} , z_{11} ; hence for x_{11} we have

$$x_{11} \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = \frac{1}{2} a_1 (y_2 z_3 - y_3 z_2) + (h_1 - \frac{1}{2} a_2) (y_3 z_1 - y_1 z_3) + (g_1 - \frac{1}{2} a_3) (y_1 z_2 - y_2 z_1),$$

or, what is the same thing,

$$L x_{11} = \frac{1}{2} a_1 \xi_1 + (h_1 - \frac{1}{2} a_2) \xi_2 + (g_1 - \frac{1}{2} a_3) \xi_3,$$

and so for y_{11} , z_{11} ; or if (as in the sequel) we desire the value of a linear function $\alpha x_{11} + \beta y_{11} + \gamma z_{11}$, calling this for a moment $\square$, we join to the foregoing a new equation

$$\alpha x_1 + \beta y_1 + \gamma z_1 = \square,$$

and then, eliminating the three quantities, we have

$$\begin{vmatrix} \alpha & \beta & \gamma & \square \\ x_1 & y_1 & z_1 & \frac{1}{2} a_1 \\ x_2 & y_2 & z_2 & h_1 - \frac{1}{2} a_2 \\ x_3 & y_3 & z_3 & g_1 - \frac{1}{2} a_3 \end{vmatrix} = 0,$$

giving $L \square$ as a linear function of $\frac{1}{2} a_1$, $h_1 - \frac{1}{2} a_2$, $g_1 - \frac{1}{2} a_3$.

We can in like manner form the expressions for the second differential coefficients of the a, b, c, f, g, h: these will of course contain third differential coefficients of the x , y , z .

Writing down only what is wanted, we have

$$\begin{aligned} \frac{1}{2} a_{22} &= 12.12 + 1.122, \\ \frac{1}{2} b_{11} &= 12.12 + 2.112, \\ h_{12} &= 12.12 + 11.22 + 1.122 + 2.112, \end{aligned}$$

where of course 12.12 denotes $x_{12}^2 + y_{12}^2 + z_{12}^2$, 1.122 denotes $x_1 \cdot x_{122} + y_1 \cdot y_{122} + z_1 \cdot z_{122}$, and so in other cases: it follows that

$$\frac{1}{2} (a_{22} + b_{11} - 2h_{12}) = 12.12 - 11.22,$$

so that the third differential coefficients of x , y , z , which enter into the expression of $a_{22} + b_{11} - 2h_{12}$ destroy each other, and this combination contains really only second differential coefficients of x , y , z .

Similarly

$$\begin{aligned} f_{12} &= 31.23 + 33.12 + 3.123 + 2.133, \\ g_{31} &= 31.23 + 33.12 + 3.123 + 1.233, \\ c_{12} &= 2 \cdot 31.23 + 2 \cdot 3.123, \\ h_{33} &= 2 \cdot 31.23 + 1.233 + 2.133, \end{aligned}$$

and thence

$$f_{12} + g_{31} - c_{12} - h_{33} = -2(31.23 - 33.12),$$

so that here again we have a combination containing only second differential coefficients of x, y, z .

There are thus, in all, the six combinations

$$\begin{aligned} b_{33} + c_{22} - 2f_{23} & \dots\dots\dots (\mathfrak{A}), \\ c_{11} + a_{33} - 2g_{31} & \dots\dots\dots (\mathfrak{B}), \\ a_{22} + b_{11} - 2h_{12} & \dots\dots\dots (\mathfrak{C}), \\ g_{23} + h_{32} - a_{23} - f_{11} & \dots\dots\dots (f), \\ h_{31} + f_{13} - b_{31} - g_{22} & \dots\dots\dots (g), \\ f_{12} + g_{21} - c_{12} - h_{33} & \dots\dots\dots (h), \end{aligned}$$

each really containing only second differential coefficients of x, y, z ; and we thus understand how each of these combinations may be expressible in terms of the first differential coefficients of (a, b, c, f, g, h) . We have, in fact, for thus expressing these combinations the six equations called $(\mathfrak{A})$, $(\mathfrak{B})$, $(\mathfrak{C})$, (f) , (g) , (h) about to be obtained, and which are the generalisations of Warren's six equations for normal coordinates.

I consider the several determinants of the form

$$\begin{vmatrix} x_{\alpha\beta} & y_{\alpha\beta} & z_{\alpha\beta} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix},$$

in all 18, since the suffixes for the top row may be 11, 22, 33, 23, 31, 12, and those for the second and third rows 2, 3; 3, 1; or 1, 2. Each determinant is a linear function of second differential coefficients of x, y, z (thus the determinant written down is $\xi_1 x_{11} + \eta_1 y_{11} + \zeta_1 z_{11}$, and as such it can, by what precedes, be expressed by means of the first differential coefficients of the a, b, c, f, g, h . Thus, if the determinant above written down be called D , writing ξ_3, η_3, ζ_3 , for α, β, γ , we have

$$\begin{vmatrix} \xi_3 & \eta_3 & \zeta_3 & \square \\ x_1 & y_1 & z_1 & \frac{1}{2}a_1 \\ x_2 & y_2 & z_2 & h_1 - \frac{1}{2}a_2 \\ x_3 & y_3 & z_3 & g_1 - \frac{1}{2}a_3 \end{vmatrix} = 0,$$

that is,

$$L \square = - \begin{vmatrix} & \xi_3 & \eta_3 & \zeta_3 \\ \frac{1}{2}a_1 & x_1 & y_1 & z_1 \\ h_1 - \frac{1}{2}a_2 & x_2 & y_2 & z_2 \\ g_1 - \frac{1}{2}a_3 & x_3 & y_3 & z_3 \end{vmatrix},$$